Robotic Systems-2026

LAB 3 REPORT

Attitude Dynamics

Team members:

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1. Introduction

In this lab, we will learn how to simulate the attitude **dynamics** using quaternions. We will recall the Euler equation and implement it for simulation of a single rigid body.

Fill in this report and **submit it to Moodle**.

2. Simulation

<u>Tasks</u>:

2.1. Write a function that describes the dynamics of a single rigid body using two equations. First is the quaternion kinematic equation obtained from previous practice and second is the Euler equation for rotational motion.

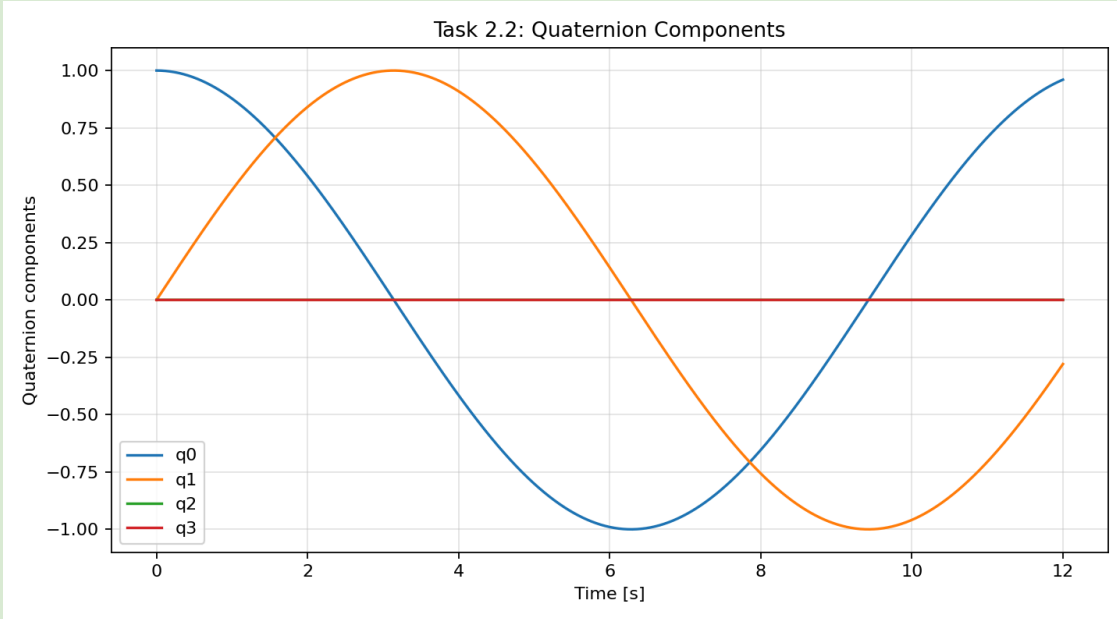
$$\begin{cases} \dot{\mathbf{q}} = \frac{1}{2}\mathcal{R}(\mathbf{q})\boldsymbol{\omega} \\ \mathbf{J}\dot{\boldsymbol{\omega}} + \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} = \boldsymbol{\tau}^B \end{cases}$$

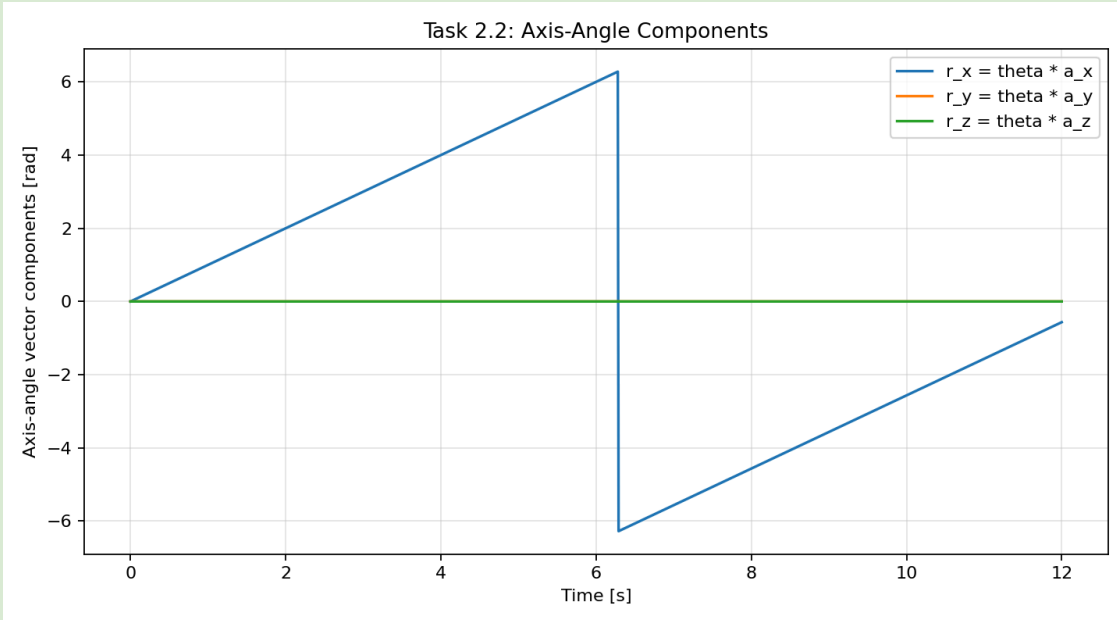
$$\mathcal{R}(\mathbf{q}) = \begin{bmatrix} -q_1 & -q_2 & -q_3 \\ q_0 & q_3 & -q_2 \\ -q_3 & q_0 & q_1 \\ q_2 & -q_1 & q_0 \end{bmatrix}$$

Present a system in state space form.

2.2. Implement Runge-Kutta method to simulate the system in state space form obtained in 2.1 (assume the torque is zero). Play with different initial angular velocities (as an example, you may take initial angular velocity as $\omega_0 = [1, 0, 0]$, $\omega_0 = [0, 1, 0]$, $\omega_0 = [0, 0, 1]$). Include the plots (for one certain initial angular velocity!) of:

- quaternion components;
- components of axis-angle representation.





Plots for Task 2.2

Explain the obtained result.

Answer(s):

Simulation was done with RK4, τ_B=0, J=diag(2,1,0.5).

Tested initial conditions: ω_0=[1,0,0], ω_0=[0,1,0], ω_0=[0,0,1].

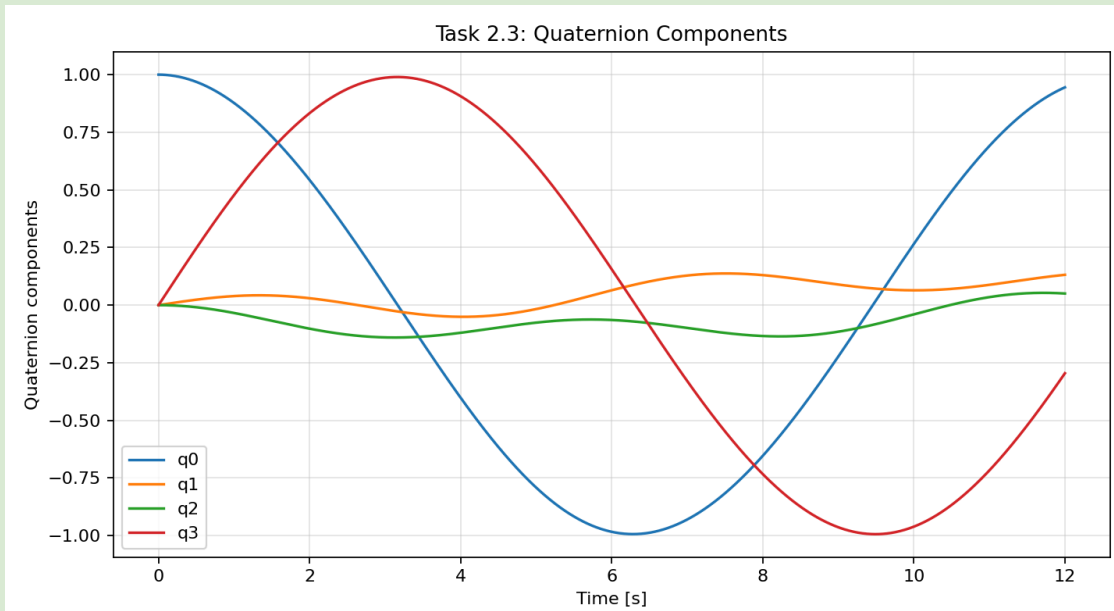
For these principal-axis initial velocities, ω remains constant, so rotation is uniform.

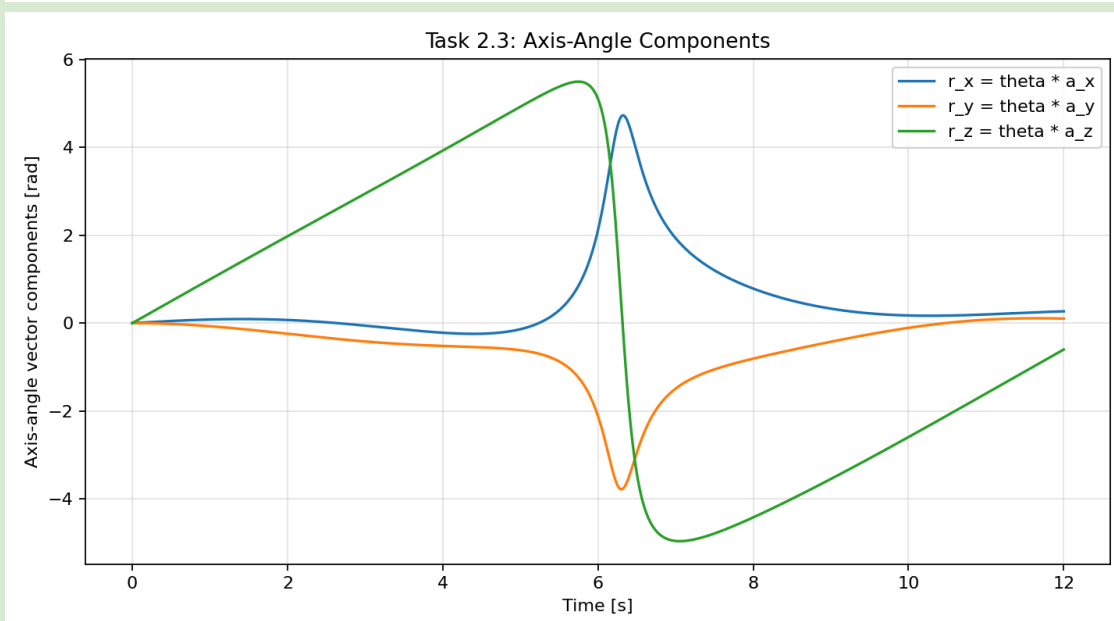
For the shown case $\omega_0=[1,0,0]$, only q_0 ,q_1 change; q_2,$q_3=0$.

Axis-angle has only the x component nonzero; the jump near $\theta=\pi$ is a wrap effect.

2.3. Change the initial angular velocity by adding some small deviation in the other component (for example, put $\omega_0 = [0.1,\ 0.,1.]$). Include the plots of:

- quaternion components;
- Components of axis-angle representation.





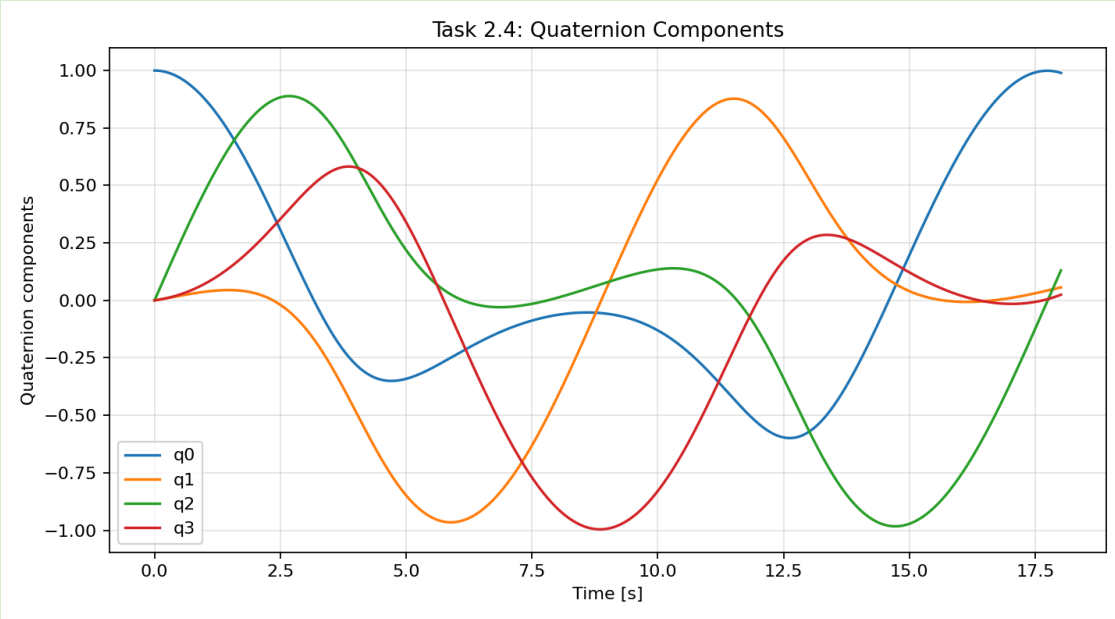
Plots for Task 2.3

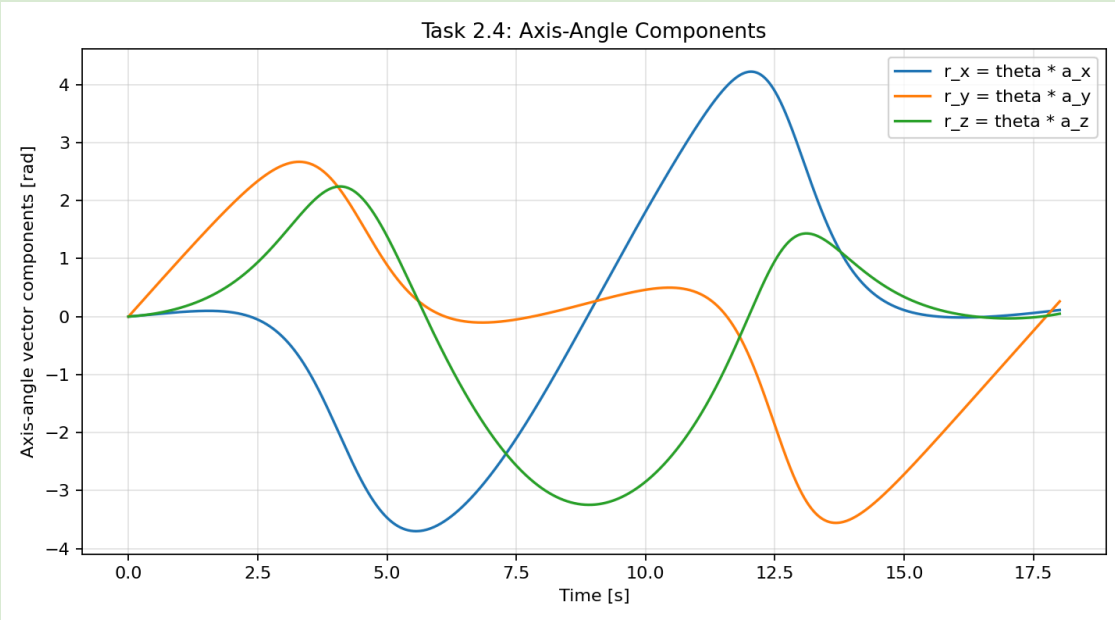
Explain the obtained result.

Answer(s):
The initial angular velocity was set to w0 = [0.1, 0, 1], which means the body spins mostly around the z-axis with a small disturbance in the x-axis. Because of this perturbation, the motion is no longer a pure single-axis rotation. The quaternion plot shows that all components become involved in the motion: q3 and q0 are still dominant, but q1 and q2 are now nonzero and oscillate. This indicates coupled rotational dynamics between axes.
In the axis-angle plot, all three components vary with time instead of only one. This is expected, because the small deviation causes the rotation axis to move (precess) rather than remain perfectly fixed. The sharp transition in the axis-angle curves is a representation effect (axis-angle wrapping to an equivalent form), not a real physical jump in orientation.
The final angular velocity differs from the initial one, which confirms cross-axis coupling under Euler rotational dynamics for a non-symmetric inertia matrix. Since external torque is zero, the behavior comes from internal rigid-body dynamics, not from external forcing.

2.4. Check the middle-axis theorem by setting some value to the y-component of initial angular velocity and including some instability in other components. Include the plots of:

- quaternion components;
- Components of axis-angle representation.





Plots for Task 2.4

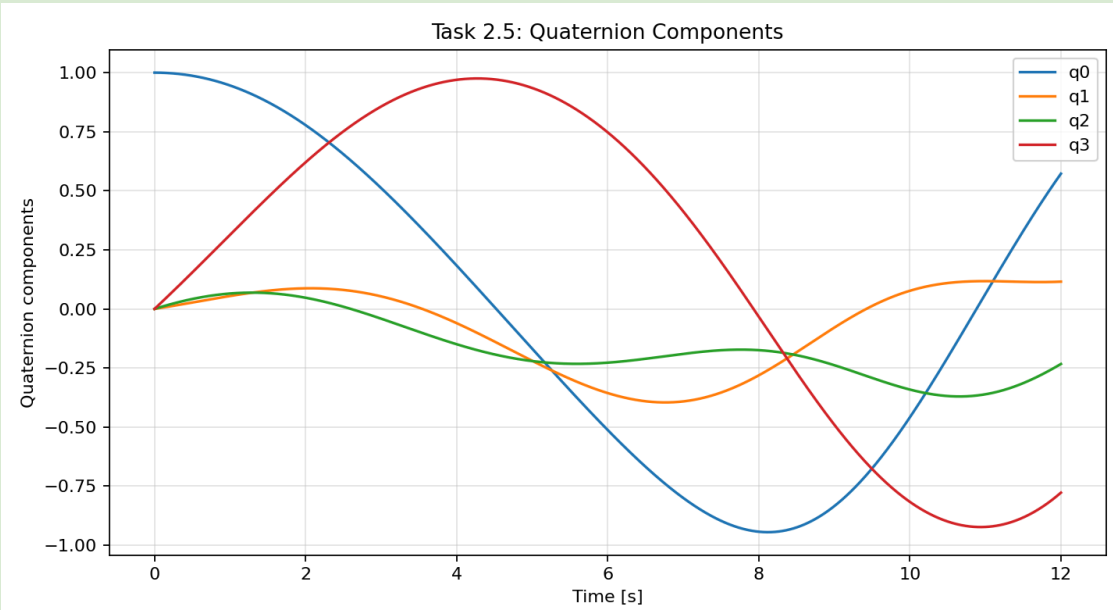
Explain the obtained result.

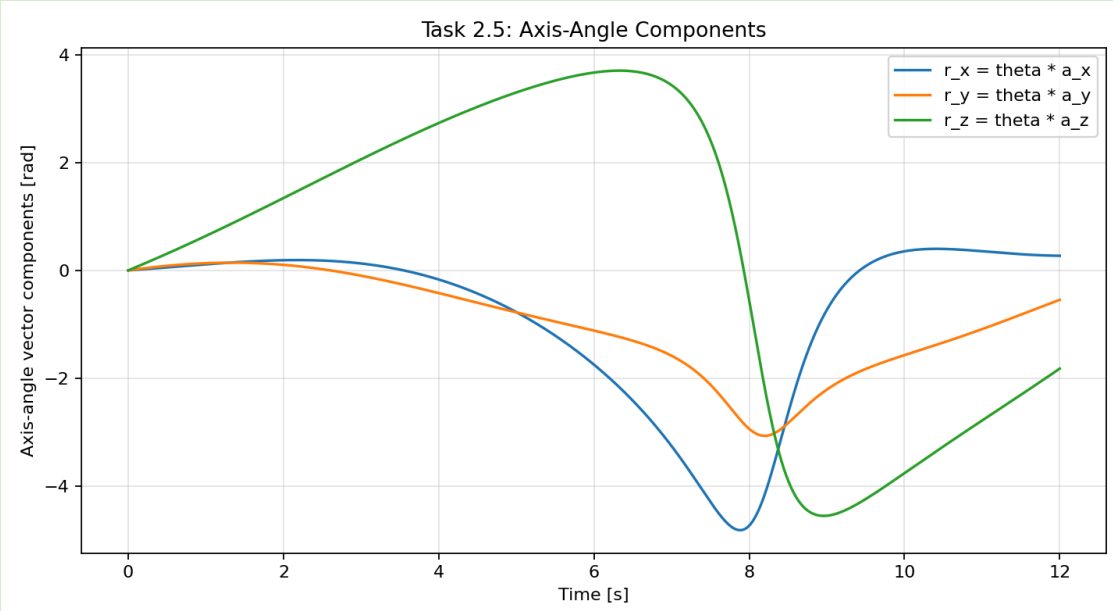
Answer(s):
Initial angular velocity was set mostly along the intermediate axis (y) with small x and z perturbations. The motion becomes unstable: the quaternion and axis-angle components show strong oscillations and redistribution between axes.

This is the expected intermediate-axis instability (tennis-racket effect), where small disturbances grow and cause attitude flipping behavior.

2.5. Set some value to the torque (this will indicate some external force applied to the system). Include the plots of:

- quaternion components;
- Components of axis-angle representation.





Plots for Task 2.5

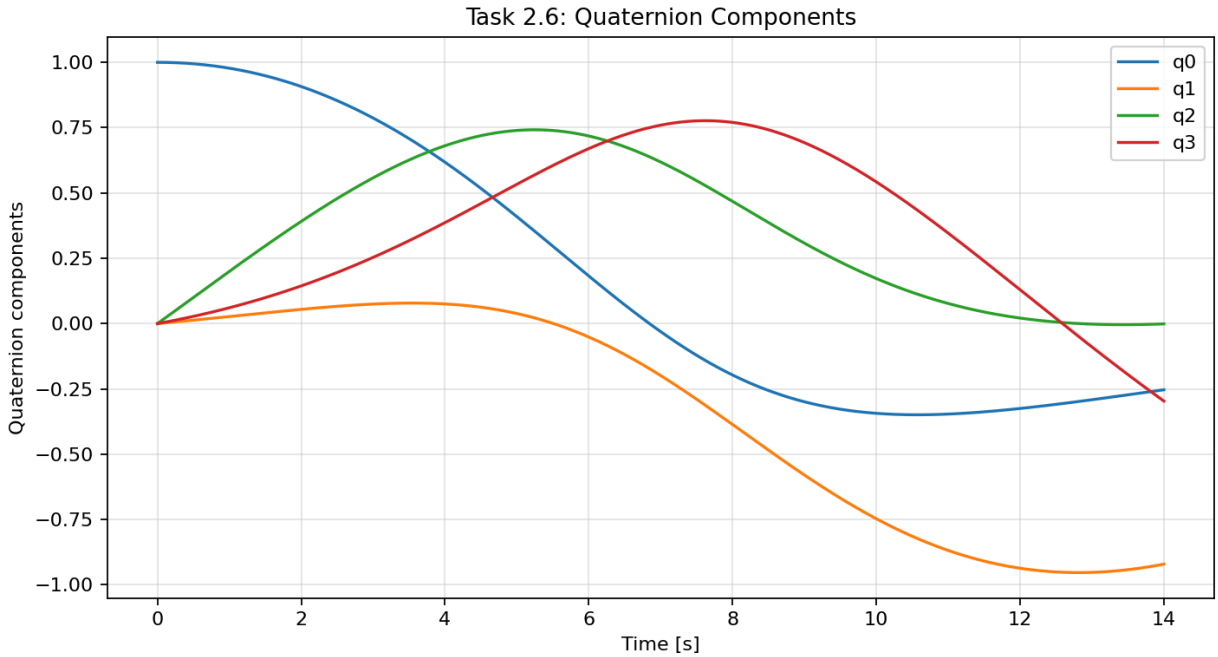
Explain the obtained result.

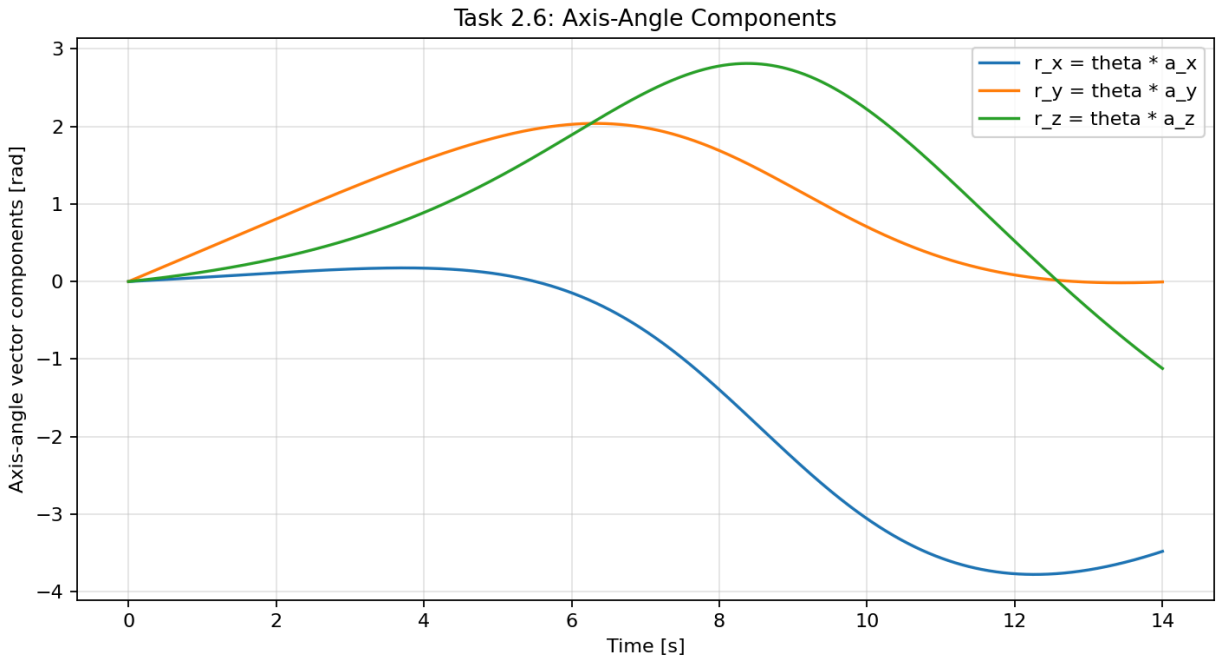
Answer(s):
A constant external torque was applied, so angular velocity is no longer conserved. Quaternion and axis-angle curves become biased and non-symmetric compared to torque-free motion. The system is continuously driven by the torque, so orientation changes faster in some directions and the final angular velocity differs significantly from the initial one.

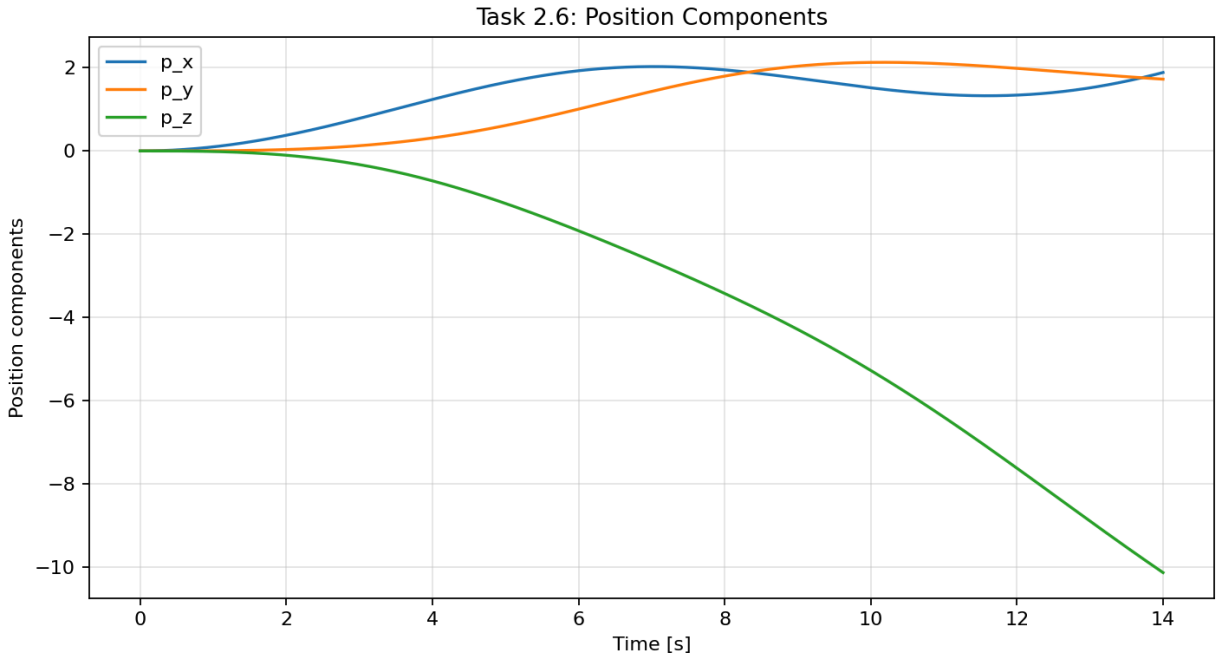
2.6. [BONUS] Now we learnt how to simulate the attitude dynamics. The next step is to include the dynamics for positional motion. Include the corresponding equations and change the state vector accordingly and simulate the system. Provide the plots of:

- quaternion components;
- Components of axis-angle representation.
- Position components

Explain the obtained result.


Task 2.6: Quaternion Components
Quaternion components
Time [s]
q0
q1
q2
q3
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
2
4
6
8
10
12
14


Task 2.6: Axis-Angle Components
Axis-angle vector components [rad]
Time [s]
r_x = theta * a_x
r_y = theta * a_y
r_z = theta * a_z
3
2
1
0
−1
−2
−3
−4
0
2
4
6
8
10
12
14



Answer(s):
The state was extended to include position and velocity:
x = [q0, q1, q2, q3, wx, wy, wz, px, py, pz, vx, vy, vz].
The added equations were:
p_dot = v,
v_dot = (1/m) * C(q) * F_body,
together with the same attitude dynamics from previous tasks.
Result: quaternion and axis-angle evolve with rotational dynamics, while position components change over time due to translational acceleration. This confirms the coupled attitude-position simulation is working correctly.

APPENDIX
In Python you may find specific libraries that allow you to go from one attitude presentation to another. Example:

```
from scipy.spatial.transform import Rotation
# transfer from Euler angles to quaternion presentation
quat_0 = Rotation.from_euler('xyz', [15, 10, 5],
degrees=True).as_quat()
```